MARKING SCHEME

Model Paper — 02

Third Term Test (Full Syllabus)

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Note (for teachers):

- Part A: 2 marks for each question (generally 1 mark for the correct method and 1 mark for the correct final answer). — Total 40 marks
- Part B: 10 marks for each question. — Total 60 marks. The marks for each step are shown in the right-hand column.
- Award full marks for correct alternative methods. Accept a reasonable range for values read from graphs.

Part A — Answers

1. $\sqrt{5.29} = \sqrt{\frac{529}{100}} = \frac{23}{10}$

Final answer: 2.3

(2 marks)

2. $\frac{3}{5} \times \frac{5}{2} = \frac{3}{2}$; $\frac{1}{4} \times 2 = \frac{1}{2}$; $\frac{3}{2} - \frac{1}{2}$

Final answer: 1

(2 marks)

3. $x^2 - 7x + 4x - 28$

Final answer: $x^2 - 3x - 28$

(2 marks)

4. $(3a)^2 - (4b)^2$

Final answer: $(3a - 4b)(3a + 4b)$

(2 marks)

5. $2(x^2 - 4x - 21)$

Final answer: $2(x - 7)(x + 3)$

(2 marks)

6. $\frac{(x - 3)(x + 3)}{(x + 2)(x + 3)}$

Final answer: $\frac{x - 3}{x + 2}$

(2 marks)

7. $\frac{90}{360} \times \frac{22}{7} \times 14 \times 14$

Final answer: 154 cm^2

(2 marks)

8. Exterior angle = sum of the two interior opposite angles ; $125° - 48°$

Final answer: 77°

(2 marks)

9. $\lg \frac{50 \times 4}{2} = \lg 100$

Final answer: 2

(2 marks)

10. $\lg 8 = 3 \lg 2 = 0.9030$; $\lg 0.8 = \lg 8 - \lg 10 = 0.9030 - 1$

Final answer: $\bar{1}.9030$

(2 marks)

11. $3 + (n - 1)4 = 87 \Rightarrow 4(n - 1) = 84 \Rightarrow n - 1 = 21$

Final answer: 22nd term

(2 marks)

12. $A' = \{1, 3, 5, 7, 9\}$; $B = \{3, 6, 9\}$

Final answer: $A' \cap B = \{3, 9\}$

(2 marks)

13. 1st year: $40\,000 \times 1.1 = 44\,000$; 2nd year: $44\,000 \times 1.1$

Final answer: Rs. 48 400

(2 marks)

14. Work done in one day = $\frac{1}{6} + \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$

Final answer: 4 days

(2 marks)

15. Time = $2\frac{1}{4}$ h = $\frac{9}{4}$ h ; $180 \div \frac{9}{4} = 180 \times \frac{4}{9}$

Final answer: 80 km/h

(2 marks)

16. $2S = n(a + l) \Rightarrow a + l = \frac{2S}{n}$

Final answer: $l = \frac{2S}{n} - a$ (or $\frac{2S - an}{n}$)

(2 marks)

17. Sample space = {HH, HT, TH, TT} ; Outcomes with at least one head = 3

Final answer: $\frac{3}{4}$

(2 marks)

18. AB = CD $\Rightarrow 2x + 3 = 15 - x \Rightarrow x = 4$; AB = 2(4) + 3

Final answer: 11 cm

(2 marks)

19. Triangle = $\frac{1}{2}$ × parallelogram = $\frac{1}{2}$ × 48

Final answer: 24 cm^2

(2 marks)

20. 3.5 × 50 000 = 175 000 cm

Final answer: 1.75 km

(2 marks)

Part B — Answers and Mark Allocation

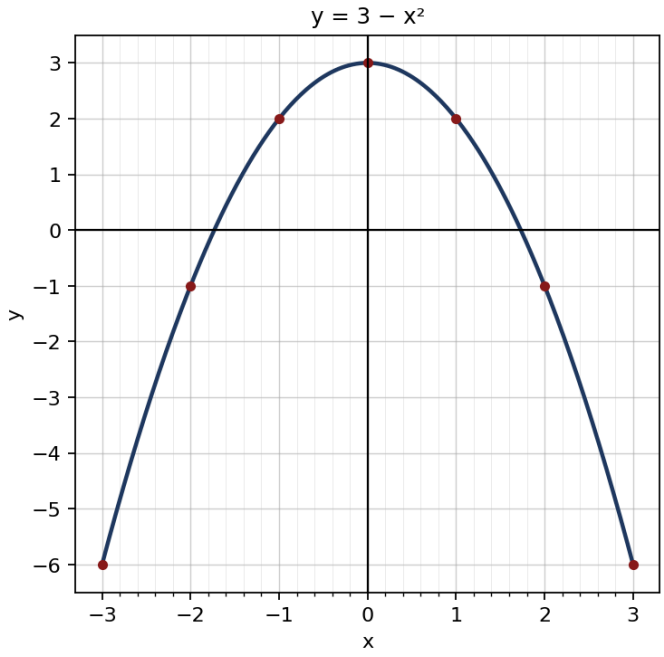
Question 1

(a)	
Common denominator $(x + 3)(x - 3)$	**1**
$\frac{2(x - 3) + 3(x + 3) - 18}{(x + 3)(x - 3)}$	**1**
$= \frac{5x - 15}{(x + 3)(x - 3)} = \frac{5(x - 3)}{(x + 3)(x - 3)}$	**1**
$= \frac{5}{x + 3}$	**1**
(b)	
Multiplying by 12: $4(x + 1) - 3(x - 2) = 24$	**1**
$4x + 4 - 3x + 6 = 24$	**1**
$x = 14$	**1**
(c)	
$2x^2 - x - 15 = 0$	**1**
$(2x + 5)(x - 3) = 0$	**1**
$x = 3$ or $x = -\frac{5}{2}$	**1**

Total marks: 10

Question 2

(a)	
$x = -3 \Rightarrow y = -6$; $x = 2 \Rightarrow y = -1$	**1**
(b)	
Correct scale and axes	**1**
All 7 points plotted correctly	**1**
Smooth curve	**1**



(c)	
(i) Maximum value 3 ; turning point (0, 3)	**1**
(ii) $-1.7 < x < 1.7$ (accept ±1.7 – ±1.8)	**1**
(iii) $3 - x^2 = 0 \Rightarrow x^2 = 3 \Rightarrow x = \pm\sqrt{3}$; the points where the curve cuts the x-axis	**1**
$\sqrt{3} \approx 1.7$	**1**
(iv) $y \approx -3.25$ (accept −3.2 – −3.3)	**1**
(d)	
$y = 1 - x^2$; (−1, 0) and (1, 0)	**1**

Total marks: 10

Question 3

(a)	

$\frac{90}{360} \times \frac{22}{7} \times 14 \times 14$	**1**
$= 154\ \text{cm}^2$	**1**
(b)	
Area of the square $= 14 \times 14 = 196\ \text{cm}^2$	**1**
Shaded region $= 196 - 154 = 42\ \text{cm}^2$	**1**
(c)	
Length of arc BD $= \frac{90}{360} \times 2 \times \frac{22}{7} \times 14$	**1**
$= 22$ cm	**1**
Perimeter $= 22 + 14 + 14 = 50$ cm	**1**
(d)	
Radius of the semicircle $= 7$ cm	**1**
Area $= \frac{1}{2} \times \frac{22}{7} \times 7 \times 7 = 77\ \text{cm}^2$	**1**
Total area $= 196 + 77 = 273\ \text{cm}^2$	**1**

Total marks: 10

Question 4

(a) In triangles ABD and ACD,	
AB = AC (given) ; BD = DC (D is the midpoint)	**1**
AD is the common side	**1**
$\therefore \Delta ABD \equiv \Delta ACD$ (S.S.S.)	**1**
(b)	
$A\hat{D}B = A\hat{D}C$ (corresponding angles of congruent triangles)	**1**
$A\hat{D}B + A\hat{D}C = 180°$ (angles on a straight line) $\Rightarrow$ each angle is $90° \Rightarrow AD \perp BC$	**1**
(c)	
$B\hat{A}D = C\hat{A}D$ (corresponding angles) $\Rightarrow$ AD bisects $B\hat{A}C$	**1**
(d)	
BD = 5 cm ; $AD^2 = 13^2 - 5^2 = 144$ (Pythagoras' theorem)	**1**
AD = 12 cm	**1**
Area = $\frac{1}{2} \times 10 \times 12$	**1**
= 60 cm^2	**1**

Total marks: 10

Question 5

(a)	
$50\,000 \times \frac{8}{100}$	**1**
= Rs. 4 000	**1**
(b)	
Interest for the 2nd year = $54\,000 \times \frac{8}{100}$ = Rs. 4 320	**1**
Total amount = Rs. 58 320	**1**
(c)	
Remaining amount = 58 320 − 18 320 = Rs. 40 000	**1**
Interest = $40\,000 \times \frac{8}{100}$ = Rs. 3 200	**1**
At the end of the 3rd year = Rs. 43 200	**1**
(d)	
Simple interest = $50\,000 \times \frac{9}{100} \times 2$ = Rs. 9 000	**1**
Compound interest (8%, 2 years) = 58 320 − 50 000 = Rs. 8 320	**1**
The 9% simple interest method is more advantageous by Rs. 680	**1**

Total marks: 10

Question 6

(a)	
$a = 500, d = 150$; $T_{10} = 500 + 9 \times 150$	**1**
= Rs. 1 850	**1**
(b)	
$S_{12} = \frac{12}{2}[2 \times 500 + 11 \times 150]$	**1**
$= 6 \times 2\,650$	**1**
= Rs. 15 900	**1**
(c)	
$500 + (n - 1)150 > 3000$	**1**
$(n - 1) > \frac{2500}{150} \approx 16.67$	**1**
$n - 1 = 17 \Rightarrow n = 18$; 18th month ($T_{17} = 2\,900$, $T_{18} = 3\,050$)	**1**
(d)	
$S_{18} = \frac{18}{2}[1000 + 17 \times 150] = 9 \times 3\,550$	**1**
= Rs. 31 950	**1**

Total marks: 10